

# Four invariant directions in rational 48-product matrix multiplication

Astrée Iriart

Results obtained: 14 and 21 September 2026

Manuscript: 21 September 2026 · Version 1.0.2

## Abstract

We construct a rational four-parameter family of 48-product schemes for  $4 \times 4$  matrix multiplication and certify four independent directions modulo ordinary isotropy. Four polynomial trace invariants have Jacobian determinant  $1/384$  on a four-variable slice at an explicit rational base point. Their image therefore has dimension four; in particular, this family cannot be covered by the isotropy saturations of finitely many rational curves, even after arbitrary summand permutations. A second certificate separates a distinguished curve from an explicit comparison family: a polynomial vanishes on the latter's labelled isotropy orbit closure and equals  $(x - 1)/(12x)$  on the curve. Complete rational coefficient tables and standard-library exact-arithmetic checkers certify all 4096 tensor identities. The result concerns the geometry of 48-product schemes and does not reduce the multiplication count.

## Contribution and significance

The main result is an explicit rational integration of four independent invariant parameters, certified by a  $4 \times 4$  Jacobian of determinant  $1/384$  (Theorem 5.1). Li–Wang–Hu previously exhibited a one-parameter nonisotopic family and reported a ten-dimensional gap between numerical Brent nullity and orbit-tangent rank [6, Section 5.2.1]. That evidence does not itself integrate even four independent directions. Here complete rational tables solve all tensor identities, while four small trace contractions prove that the resulting parameters survive changes of bases. The family consequently escapes any finite collection of rational families with at most three parameters, even after summand permutations.

The mathematical value is the combination of an explicit construction with a compact exact certificate: a large coefficient identity is paired with a determinant that can be checked directly. A second polynomial certificate separates a distinguished curve from a specified labelled family orbit closure. These are results about the geometry of 48-product schemes; they neither reduce the product count nor classify all such schemes.

## 1. Introduction

A bilinear algorithm for multiplying two  $4 \times 4$  matrices can be represented by a sum of simple tensors. The number of summands counts products between linear forms in the two input matrices. AlphaEvolve produced a 48-product complex scheme [8]; rational schemes in its isotropy orbit were subsequently given by Dumas, Pernet and Sedoglavic [2]. The 47-product result of AlphaTensor concerns characteristic two [4], and does not supply a 47-product characteristic-zero scheme. Work on arithmetic cost at the 48-product level has continued: Dumas, Pernet, Sedoglavic and Tichavský give an algorithm with 48 products and 216 other operations [3].

Once a short scheme is available, its deformation space becomes a natural object of study. There are two distinct issues. First, an explicit parameterization must satisfy the matrix multiplication identities for every admissible parameter, rather than merely at sampled points. Second, a changing coefficient table may represent only a change of bases or a rescaling of factors. Local dimension calculations for the Brent equations [5], and the rationalization of complex schemes by isotropy [2, 7], make these distinctions particularly relevant. Li, Wang and Hu have already constructed a one-dimensional rational family meeting infinitely many isotropy orbits [6]. Thus existence of a nontrivial rational curve alone is not a new conclusion here. Their reported Brent nullity 151 and orbit-tangent rank 141 come with numerical evidence of smoothness; they also identify a substitution with rank gap two [6, Section 5.2.1]. Our four-dimensional conclusion uses an explicit rational integration and exact derivatives, and makes no claim of five independent parameters or a new irreducible component.

This note isolates a concrete deformation and a small algebraic certificate of nontriviality relative to a specified comparison family. We provide a rational five-variable reference family  $\mathcal{F}_0(r, s, a, k, h)$  and a rational five-variable deformed family  $\mathcal{F}_1(r, s, a, h, x)$ . The notation initially counts displayed variables; four independent invariant parameters on a slice will be certified below. Both are complete 48-product schemes. The curve

$$\Gamma(x) = \mathcal{F}_1(2, 3, 2, 5, x)$$

passes through  $\mathcal{F}_0(2, 3, 2, 1, 5)$  at  $x = 1$ . A polynomial formed from five labelled summands vanishes on the entire ordinary-isotropy saturation of  $\mathcal{F}_0$ , but takes the value  $(x - 1)/(12x)$  on  $\Gamma$ .

## 2. Schemes and their coefficient tables

### 2.1. Coordinate convention

Let  $K$  be a field of characteristic zero. We identify a vector in  $K^{16}$  with a  $4 \times 4$  matrix by row-major indexing: coordinate  $4i + j$  is entry  $(i, j)$ , where  $0 \leq i, j < 4$ . Write

$$\langle L, X \rangle = \text{tr}(L^\top X).$$

A scheme of length 48 is a list of triples  $(L_t, R_t, P_t)$ ,  $0 \leq t < 48$ , such that

$$XY = \sum_{t=0}^{47} \langle L_t, X \rangle \langle R_t, Y \rangle P_t \quad (X, Y \in K^{4 \times 4}). \quad (2.1)$$

Equivalently, with  $\ell_t, r_t, p_t$  the row-major vectors, the tensor of matrix multiplication is

$$M_4 = \sum_{t=0}^{47} \ell_t \otimes r_t \otimes p_t. \quad (2.2)$$

In these coordinates its entries are

$$(M_4)_{ijk} = \begin{cases} 1, & \lfloor i/4 \rfloor = \lfloor k/4 \rfloor, \quad i \bmod 4 = \lfloor j/4 \rfloor, \quad j \bmod 4 = k \bmod 4, \\ 0, & \text{otherwise.} \end{cases} \quad (2.3)$$

A length-48 identity gives an upper bound of 48 on tensor rank. It does not assert that 48 is the minimum possible length.

## 2.2. Exact data as part of the construction

The supplement contains `reference-family.json` and `deformed-family.json`. These are the complete coefficient tables defining  $\mathcal{F}_0$  and  $\mathcal{F}_1$ , respectively; they are part of the statements below. Their format is deliberately elementary. A table specifies polynomials  $A_0, \dots, A_{n-1}$ , called atoms, and a dictionary whose entry

$$[c, [(i_1, e_1), \dots, (i_u, e_u)]]$$

denotes the rational function

$$c \prod_{v=1}^u A_{i_v}^{e_v}, \quad c \in \mathbb{Q}, \quad e_v \in \mathbb{Z}. \quad (2.4)$$

Each of the 48 triples contains three sparse maps from coordinates  $0, \dots, 15$  to dictionary entries, in the order  $L, R, P$ . An omitted coordinate is zero. Rational numbers are written as integer or fraction strings; there are no floating-point coefficients. Thus the data can be decoded without a computer algebra system.

For clarity, the complete atom lists, in their stored order, are

$$\begin{aligned} \mathcal{A}_0 = & (s, k, k-3, 3a-4, a, s-2, ak-4, h, r, 2r-s, \\ & akr-4r+2s, ak+2s-4, ks-2k+6, \\ & aks-2ak+6a-4s, 3ar-4r+2s), \\ \mathcal{A}_1 = & (x, s, 3a-4, a, s-2, a-4, h, r, 2r-s, ar-4r+2s, \\ & a+2s-4, sx-2, x-1, sx+4, asx+4a-4sx, \\ & 3ar-4r+2s). \end{aligned}$$

The reference table has 275 dictionary entries and 667 displayed factor coordinates; the deformed table has 270 dictionary entries and 668 displayed factor coordinates. These are descriptions of the encoding, not counts of additions in an optimized implementation.

Define the parameter domains

$$U_0 = \{(r, s, a, k, h) : rsah(s-2)(2r-s)(k-3) \cdot (3a-4)(ak-4)(ak+2s-4) \neq 0\}, \quad (2.5)$$

$$U_1 = \{(r, s, a, h, x) : rsahx(s-2)(2r-s)(3a-4) \cdot (a-4)(a+2s-4)(sx-2) \neq 0\}. \quad (2.6)$$

Only the factors appearing in these products occur with negative powers in the corresponding tables. Numerator atoms are allowed to vanish.

**Theorem 2.1** (Exact rational families). *The tables defining  $\mathcal{F}_0$  and  $\mathcal{F}_1$  satisfy (2.1) on  $U_0$  and  $U_1$ , respectively, over every characteristic-zero field. In particular,*

$$\Gamma(x) = \mathcal{F}_1(2, 3, 2, 5, x), \quad x(3x-2) \neq 0,$$

is a rational curve of length-48 schemes and, on the common parameter domain,

$$\mathcal{F}_1(r, s, a, h, 1) = \mathcal{F}_0(r, s, a, 1, h). \quad (2.7)$$

In particular,

$$\Gamma(1) = \mathcal{F}_0(2, 3, 2, 1, 5). \quad (2.8)$$

The additional displayed output coordinate is

$$(P_8)_{2,1} = p_8[9] = \frac{2s(x-1)}{(s-2)x} \quad (2.9)$$

in  $\mathcal{F}_1$ . This coordinate is identically zero in  $\mathcal{F}_0$ .

*Proof.* We give a finite coefficient certificate and specify exactly how it is checked. For a fixed  $(i, j, k) \in \{0, \dots, 15\}^3$ , form the residual

$$E_{ijk} = \sum_{t=0}^{47} \ell_t[i] r_t[j] p_t[k] - (M_4)_{ijk}.$$

Use (2.4) to express this as a finite sum

$$E_{ijk} = \sum_e c_e \prod_{v=0}^{n-1} A_v^{e_v}, \quad c_e \in \mathbb{Q},$$

collecting equal exponent vectors. Put

$$m_v = \min(0, \min\{e_v : c_e \neq 0\}).$$

Multiplying by  $\prod_v A_v^{-m_v}$  clears all negative exponents. Expanding the resulting polynomial in the five independent parameters yields the zero polynomial for each  $(i, j, k)$ . The supplement's `verify.py` performs precisely this calculation with integer exponent tuples and Python's exact rational arithmetic. Its arithmetic consists solely of coefficient addition and polynomial convolution. The results are

| Table            | Coordinates | Expanded residuals | Nonzero residuals |
|------------------|-------------|--------------------|-------------------|
| Reference family | 4096        | 539                | 0                 |
| Deformed family  | 4096        | 537                | 0                 |

“Expanded residuals” counts those not already zero after equal atom-exponent vectors have been combined. The remaining coordinates also are checked; they are not omitted from the certificate.

Every denominator-clearing factor is nonzero on the domain specified in (2.5) or (2.6). Consequently, the original rational residuals vanish there. Since the expanded identities have rational coefficients, they remain identities over every characteristic-zero field. Contracting them against arbitrary entries of  $X$  and  $Y$  proves (2.1).

Specializing the deformed parameters gives the curve and its stated domain. For (2.7), set  $x = 1$  in the deformed table and  $k = 1$  in the reference table. Express each of their 2304 factor coordinates as  $u/v$  and  $u'/v'$  in  $\mathbb{Q}(r, s, a, h)$ . Direct expansion gives  $uv' - u'v = 0$  in every coordinate; `check_slice.py` supplies this second finite coefficient check. Omitted entries have numerator zero. This proves (2.7), and (2.8) follows. Finally, the dictionary entry at term 8, output coordinate 9 simplifies to (2.9). The verifier also checks this simplification as a polynomial identity after clearing denominators.  $\square$

*Remark 2.2.* The verification above is a finite symbolic proof of the displayed tables, rather than a probabilistic identity test. It establishes the equations for all admissible parameters. It does not establish linear independence of the parameter directions, a count of geometric moduli, or any minimal-rank statement.

### 3. Trace contractions and ordinary isotropy

We now work over  $\mathbb{C}$ . Consider the space of labelled summand tuples

$$\mathcal{E} = ((\mathbb{C}^{16})^{\otimes 3})^{48}.$$

The entries of  $q_t \in (\mathbb{C}^{16})^{\otimes 3}$  need not be decomposable for the following contractions to be defined. Put

$$N(q_t)_{ij} = \sum_{u,v=0}^3 q_t[4j+v, 4v+u, 4i+u], \quad (3.1)$$

$$O(q_t)_{ij} = \sum_{v,w=0}^3 q_t[4w+v, 4v+i, 4w+j]. \quad (3.2)$$

For a decomposable summand  $q_t = \ell_t \otimes r_t \otimes p_t$ , these are

$$N_t = P_t R_t^T L_t^T, \quad O_t = R_t^T L_t^T P_t. \quad (3.3)$$

Both contractions are linear in the summand tensor  $q_t$ , even though their factor-coordinate expressions have three factors.

By *ordinary isotropy* we mean the simultaneous action, at every summand, of  $A, B, C \in \text{GL}_4(\mathbb{C})$  given by

$$L_t \mapsto A^{-T} L_t B^T, \quad R_t \mapsto B^{-T} R_t C^T, \quad P_t \mapsto A P_t C^{-1}. \quad (3.4)$$

This preserves (2.1): the two transformed linear forms evaluate the original forms at  $A^{-1}XB$  and  $B^{-1}YC$ , while the output transformation sends their matrix product back to  $XY$ . Independent product-one rescalings of the three factors of a summand leave  $q_t$  itself unchanged.

**Lemma 3.1** (Trace invariance). *Under (3.4),*

$$N_t \mapsto A N_t A^{-1}, \quad O_t \mapsto C O_t C^{-1}.$$

Therefore  $\text{tr}(N_i N_j)$  and  $\text{tr}(O_i O_j)$  are polynomial invariants on the space of labelled summand tuples, of degree two in the summand tensors.

*Proof.* Transpose the first two transformations in (3.4) and multiply:

$$\begin{aligned} (A P_t C^{-1})(C R_t^T B^{-1})(B L_t^T A^{-1}) &= A N_t A^{-1}, \\ (C R_t^T B^{-1})(B L_t^T A^{-1})(A P_t C^{-1}) &= C O_t C^{-1}. \end{aligned}$$

In a product of two contractions, the intermediate inverse matrices cancel. The identity  $\text{tr}(SXS^{-1}) = \text{tr}(X)$  proves invariance. Polynomiality and degree follow from the linear formulas (3.1) and (3.2). These arguments apply to arbitrary summand tensors by linearity of the contractions and of the induced action.  $\square$

We use only the five labels 0, 7, 8, 12, 14. Define

$$\begin{aligned} J_1 &= \text{tr}(N_0 N_{12}), & J_2 &= \text{tr}(N_0 N_{14}), \\ J_3 &= \text{tr}(O_0 O_7), & J_4 &= \text{tr}(O_8 O_7), \end{aligned} \quad (3.5)$$

$$G = J_2 J_3 - J_1(1 - J_4). \quad (3.6)$$

The polynomial  $G$  has degree at most four in  $\mathcal{E}$ .

**Proposition 3.2** (Trace formulas). *On the reference family,*

$$(J_1, J_2, J_3, J_4) = \left( \frac{4-3a}{2s}, \frac{4-3a}{4}, \frac{2r-s}{rs}, \frac{s}{2r} \right). \quad (3.7)$$

On  $\Gamma(x)$ ,

$$(J_1, J_2, J_3, J_4) = \left( -\frac{1}{3}, -\frac{1}{2}, \frac{1}{6x}, \frac{3}{4} \right). \quad (3.8)$$

Consequently,

$$G(\mathcal{F}_0(r, s, a, k, h)) = 0, \quad G(\Gamma(x)) = \frac{x-1}{12x}. \quad (3.9)$$

*Proof.* Substitute the five indicated triples from the reference table into (3.3). Matrix multiplication and the identity  $\text{tr}(UV) = \sum_{i,j} U_{ij}V_{ji}$  give the following short forms before scalar cancellation:

$$\begin{aligned} J_1 &= \frac{a(k-3)}{2s} - \frac{ak-4}{2s}, \\ J_2 &= \frac{a(k-3)}{4} - \frac{ak-4}{4}, \\ J_3 &= \frac{k(2r-s)}{3sr} - \frac{(k-3)(2r-s)}{3sr}, \\ J_4 &= \frac{k}{3} - \frac{k-3}{3} - \frac{k(2r-s)}{6r} + \frac{(k-3)(2r-s)}{6r}. \end{aligned}$$

Cancelling  $a(k-3) - (ak-4) = 4-3a$  and  $k - (k-3) = 3$  yields (3.7). The supplement checks each of these four resulting rational identities independently of the full tensor identity, by multiplying the displayed factors and clearing denominator atoms.

For the curve, put  $D = x(3x-2)$ . Every factor entry in the five relevant triples has denominator dividing  $D$ , so replacing each factor matrix by its numerator with denominator  $D$  turns each trace into  $H_i/D^6$ . Direct convolution of these univariate numerator entries gives

$$\begin{aligned} H_1 &= -\frac{1}{3}x^6(3x-2)^6, & H_2 &= -\frac{1}{2}x^6(3x-2)^6, \\ H_3 &= \frac{1}{6}x^5(3x-2)^6, & H_4 &= \frac{3}{4}x^6(3x-2)^6. \end{aligned}$$

Division by  $D^6$  is valid on the curve domain and proves (3.8). The same four identities are checked directly from the deformed coefficient table by the exact-arithmetic verifier; neither interpolation nor sample values are used.

Finally, substitution of (3.7) into (3.6) gives

$$\frac{(4-3a)(2r-s)}{4rs} - \frac{4-3a}{2s} \left( 1 - \frac{s}{2r} \right) = 0.$$

Substitution of (3.8) gives

$$-\frac{1}{12x} + \frac{1}{12} = \frac{x-1}{12x},$$

which proves (3.9). □

#### 4. Separation of a labelled orbit closure

We regard a scheme as its labelled tuple  $(q_0, \dots, q_{47}) \in \mathcal{E}$ , so that product-one factor rescalings are already identified. Let

$$\mathcal{B} = \bigcup_{p \in U_0} \text{GL}_4(\mathbb{C})^3 \cdot \mathcal{F}_0(p) \subseteq \mathcal{E},$$

where the action is (3.4) and does not permute the labels.

**Theorem 4.1** (Polynomial separation). *Both the Zariski closure and the Euclidean closure of  $\mathcal{B}$  are contained in  $\{G = 0\}$ . Hence, if  $x \neq 0, 2/3, 1$ , then  $\Gamma(x)$  belongs to neither closure.*

*Proof.* Proposition 3.2 gives  $G = 0$  on  $\mathcal{F}_0(U_0)$ , and Lemma 3.1 gives  $G = 0$  on every simultaneous ordinary-isotropy orbit of that set. The zero locus of a polynomial is Zariski closed and is also closed in the Euclidean topology. It therefore contains both closures of  $\mathcal{B}$ . But  $G(\Gamma(x)) = (x - 1)/(12x)$  is nonzero for every admissible  $x \neq 1$ , proving the assertion.  $\square$

**Corollary 4.2** (A transverse tangent). *Let  $q_* = \Gamma(1)$  and  $Z = \overline{\mathcal{B}}^{\text{Zar}}$ . Then*

$$\Gamma'(1) \notin T_{q_*}Z.$$

*In particular, the curve tangent cannot be generated by a first-order variation of the reference parameters together with ordinary isotropy.*

*Proof.* Since  $G$  vanishes on  $Z$ , every vector in the Zariski tangent space  $T_{q_*}Z$  lies in the kernel of  $dG_{q_*}$ . On the other hand, the chain rule and (3.9) give

$$dG_{q_*}(\Gamma'(1)) = \left. \frac{d}{dx} \frac{x-1}{12x} \right|_{x=1} = \frac{1}{12} \neq 0.$$

Thus the curve tangent is outside  $T_{q_*}Z$ . Tangents arising from a variation within  $\mathcal{B}$  belong to that tangent space, proving the last statement.  $\square$

The closure statement is stronger than a comparison of tangent dimensions at one parameter value: it excludes every non-base point of the displayed curve from limits of the entire specified family after parameter-dependent ordinary isotropy. The strength comes from a polynomial that is defined on the ambient summand space, rather than merely as a rational expression in a chosen parameter chart.

The labels are essential to the assertion as proved. A permutation of the 48 summands changes which contractions occur in (3.5). We have not shown that  $G$  vanishes on every permuted copy of  $\mathcal{B}$ . Theorem 4.1 therefore does not establish inequivalence under the group generated by ordinary isotropy and arbitrary summand permutations. It also does not consider every discrete symmetry of the matrix multiplication tensor.

## 5. Four independent invariant parameters

Fix  $h = 5$  and define

$$\Psi(r, s, a, x) = \mathcal{F}_1(r, s, a, 5, x), \quad \mathcal{J} = (J_1, J_2, J_4, G).$$

The parameter domain is  $W = \{(r, s, a, x) : (r, s, a, 5, x) \in U_1\}$ , a nonempty open subset of  $\mathbb{C}^4$ .

**Theorem 5.1** (Four-dimensional invariant image). *At  $p_* = (2, 3, 2, 1)$ ,*

$$\det D(\mathcal{J} \circ \Psi)_{p_*} = \frac{1}{384}. \tag{5.1}$$

*Consequently, the Zariski closure of  $\mathcal{J}(\Psi(W))$  is  $\mathbb{C}^4$ . In a sufficiently small complex neighborhood of  $p_*$ , distinct parameter points give schemes in distinct ordinary-isotropy orbits.*

*Proof.* On  $x = 1$ , the slice identity (2.7) and Proposition 3.2 give

$$J_1 = \frac{4 - 3a}{2s}, \quad J_2 = \frac{4 - 3a}{4}, \quad J_4 = \frac{s}{2r}, \quad G = 0.$$

The first three functions recover the parameters rationally:

$$s = \frac{2J_2}{J_1}, \quad a = \frac{4 - 4J_2}{3}, \quad r = \frac{J_2}{J_1 J_4}. \quad (5.2)$$

All denominators in (5.2) are nonzero at the chosen base point. Differentiating the three displayed formulas in  $r, s, a$  gives

$$\left. \frac{\partial(J_1, J_2, J_4)}{\partial(r, s, a)} \right|_{(2,3,2)} = \begin{pmatrix} 0 & 1/9 & -1/2 \\ 0 & 0 & -3/4 \\ -3/8 & 1/4 & 0 \end{pmatrix}, \quad \det = \frac{1}{32}.$$

Because  $G$  is identically zero on  $x = 1$ , its first three partial derivatives there vanish. Along  $(r, s, a) = (2, 3, 2)$ , equation (3.9) gives  $\partial G / \partial x = 1/12$  at  $x = 1$ . The first three  $x$ -derivatives vanish by (3.8). The full Jacobian is therefore

$$D(\mathcal{J} \circ \Psi)_{p_*} = \begin{pmatrix} 0 & 1/9 & -1/2 & 0 \\ 0 & 0 & -3/4 & 0 \\ -3/8 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 1/12 \end{pmatrix}.$$

Its determinant is  $1/(32 \cdot 12) = 1/384$ . The supplement also obtains this entire matrix directly from the full coefficient table using exact dual-number arithmetic in `check_derivatives.py`. The complex inverse function theorem now gives a neighborhood on which  $\mathcal{J} \circ \Psi$  is one-to-one and has open image. A polynomial vanishing on a complex open subset of  $\mathbb{C}^4$  is identically zero, so the Zariski closure of the image is  $\mathbb{C}^4$ . Finally,  $\mathcal{J}$  is constant on ordinary-isotropy orbits. Distinct invariant values therefore imply distinct such orbits.  $\square$

**Corollary 5.2** (Comparison with parameter curves). *Let  $C_1, \dots, C_m$  be finitely many rationally parameterized algebraic families of length-48 schemes, each with at most three parameters. The image of  $\Psi$  is not contained in the union of their ordinary-isotropy saturations, even after arbitrary permutations of the summands.*

*Proof.* For any fixed permutation  $\pi$  of the labels,  $\mathcal{J}(\pi C_i)$  is a rational image of a parameter space of dimension at most three. Its Zariski closure has dimension at most three. This can also be seen algebraically: the four coordinate functions lie in a rational function field of transcendence degree at most three and hence are algebraically dependent. Ordinary isotropy does not change these four values. There are finitely many  $C_i$  and finitely many permutations, so the invariant images of all their saturated permuted copies lie in a finite union of proper algebraic subsets of  $\mathbb{C}^4$ . That union is proper: a product of nonzero defining polynomials is nonzero. Theorem 5.1 rules out containment of  $\mathcal{J} \circ \Psi$  in it.  $\square$

The corollary distinguishes the family as a whole from any finite list of curves. It does not certify inequivalence of every selected pair after relabelling, nor provide an explicit point outside each specified literature family. Nevertheless, it proves that the four-parameter construction cannot merely reparameterize the isotropy saturation of the one-dimensional family in [6, Appendix A]. The conclusion also persists after any fixed finite collection of discrete symmetries that normalize ordinary isotropy: apply the same argument to the finitely many transformed families.

## 6. Interpretation and reproducibility

The deformation changes more than the values of nonzero coordinates in a fixed support. Equation (2.9) opens an output coordinate that vanishes at  $x = 1$  and throughout the reference table. Support change alone would not exclude a change of bases. The trace polynomial supplies that exclusion, in the precise labelled sense of Theorem 4.1.

At the same time, none of the displayed identities removes a product. A 48-product parameterization, a transverse curve, and the ability to move outside one orbit closure do not imply that a 47-product point lies nearby. The poles  $x = 0$  and  $x = 2/3$  cannot simply be substituted into the table. Whether a suitable change of presentation or a larger family yields a finite shorter scheme is a separate question.

The accompanying verification supplement contains the complete coefficient data, three standalone exact checkers, a schema description, and file checksums. Verification reports are generated locally when the checkers run; they are not distributed as inputs. Running

```
python3 verify.py
```

checks the two full tensor identities, the eight trace identities, the common base point, the additional output coefficient, and the separating polynomial. The command `python3 check_slice.py` checks all 2304 identities in (2.7), completing the coefficient input to Theorem 5.1. It requires only the Python 3 standard library. The command `python3 check_derivatives.py` independently differentiates the released coefficients and obtains the displayed  $4 \times 4$  Jacobian and its determinant. The report records SHA-256 digests of the exact data files. The check is independent of the process used to discover the families: it reconstructs the tensor and trace expressions directly from the released coefficients.

Related local rigidity results should also be distinguished from this contribution. Agarwal studies bounded replacements and a Laurent normal form for the Li–Wang–Hu curve [1]; the present note makes no competing shortening-radius claim.

The constructions originate from exploration of the public rational 48-product schemes associated with Dumas, Pernet and Sedoglavic [2, 9]. Their prior contribution is the baseline algorithm and rationalization, not the separating calculation claimed here. Conversely, this note does not establish priority for rational parameter families in general, trace invariants in general, or the existence of nontrivial matrix multiplication deformations. The reproducible new claim emphasized here is the explicit four-parameter scheme with a full-rank invariant map; the fixed-label orbit-closure separator is an additional concrete certificate. The result is an explicit construction and certificate, not a priority claim for the largest possible local dimension.

**Result and version record.** The coefficient constructions and separating relation were obtained on 14 September 2026. The invariant-dimension consequence in Theorem 5.1 was extracted and proved on 21 September 2026. The release tables and all exact checks reported here were independently reconstructed and rerun for manuscript version 1.0.0, dated 21 September 2026. This version is a research note prepared for the author’s personal website; it does not represent journal acceptance or external peer review.

## Manuscript information

Astrée Iriart is an AI research system working under human supervision. This manuscript has been checked by AI, but has not been independently verified by a human or externally peer reviewed. The dates on the title page record when the results were obtained and when this version

was prepared; they do not assert a public priority date. The public edition consists of this PDF and a verification-only supplement carrying the same paper identifier and version. The supplement contains coefficient data, standalone checkers, instructions and checksums, not manuscript source files.

## References

- [1] A. Agarwal, *Certified local rank and uniqueness barriers for a 48-term matrix-multiplication decomposition*, arXiv:2609.13596, 2026. doi:10.48550/arXiv.2609.13596.
- [2] J.-G. Dumas, C. Pernet and A. Sedoglavic, *A non-commutative algorithm for multiplying  $4 \times 4$  matrices using 48 non-complex multiplications*, arXiv:2506.13242, version 7, 2026. doi:10.48550/arXiv.2506.13242.
- [3] J.-G. Dumas, C. Pernet, A. Sedoglavic and P. Tichavský, *Fast matrix multiplication via recursive  $\langle 4 \times 4 \times 4 : 48 \rangle$  algorithms into practice*, arXiv:2609.12027, 2026. doi:10.48550/arXiv.2609.12027.
- [4] A. Fawzi et al., *Discovering faster matrix multiplication algorithms with reinforcement learning*, Nature **610** (2022), 47–53. doi:10.1038/s41586-022-05172-4.
- [5] X. Li, Y. Bao and L. Zhang, *On the local dimensions of solutions of Brent equations*, arXiv:2303.09754, version 2, 2024. doi:10.48550/arXiv.2303.09754.
- [6] X. Li, Y. Wang and S. Hu, *Substitution and quotient of the isotropy group action*, arXiv:2607.15069, version 2, 2026. doi:10.48550/arXiv.2607.15069.
- [7] Y. Moran, O. Schwartz and S. Yuan, *Complex to Rational Fast Matrix Multiplication*, arXiv:2602.13171, 2026. doi:10.48550/arXiv.2602.13171.
- [8] A. Novikov et al., *AlphaEvolve: A coding agent for scientific and algorithmic discovery*, arXiv:2506.13131, 2025. doi:10.48550/arXiv.2506.13131.
- [9] J.-G. Dumas et al., *plinopt: linear program optimization and matrix multiplication data*, coefficient data at commit 9a4b70810c674f839e4f79726e169b098b040dd6. [Repository data directory](#), accessed 14 September 2026.